Reading Time: An initial **2 minutes** to view **BOTH** sections

MATHEMATICS METHODS : UNITS 1 & 2, 2023

Test 1 – (10%)

1.3.1. to 1.3.7, 1.2.19 to 1.2.21, 1.2.24
1.2.1 to 1.2.8, 1.2.13

PM

Time Allowed	First Name	Surname	Marks
28 minutes			28 marks

Circle your Teacher's Name: Mrs Alvaro Mr Bartoshefski Mr Buckland Mr Liu Mr Lockyer Mr Luzuk Mr Mohammadi Ms Narendranathan Mr Ng Mr Stephens Mr Tanday

Assessment Conditions: ***(N.B. Sufficient working out must be shown to gain full marks)***

- Calculators: Not Allowed
- Formula Sheet: Provided
- Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 [4 marks – 2,2]

If $f(x)=2x+5$ and $g(x)=x^2-1$, find;

(a) $f(x^2)-g(x)$

$$=(2x^2+5)-(x^2-1)$$
$$=x^2+6$$

(b) k , such that $f(-2k)=g(0)$

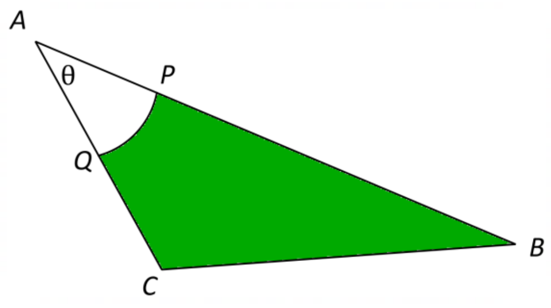
$$-4k+5=-1$$
$$\Rightarrow \quad k=\frac{3}{2}$$

Question 2 [4 marks]

Function	$y=3-x$	$y=-(x-3)^2-4$	$y=\dfrac{1}{x+5}+3$	$y=\sqrt{x-8}$
Domain	$2<x\leq 7$	$x\in R$ or $(-\infty,\infty)$ or $-\infty<x<\infty$	$x\neq -5$	$x\geq 8$
Range	$-4\leq y<1$	$y\leq -4$	$y\neq 3$	$y\geq 0$

Question 3 [4 marks – 2,2]

Triangle ABC has $AB = 8cm$ and $AC = 5cm$. A circle, centre A and radius $3cm$ intersects AB and AC at P and Q respectively. The area of triangle ABC is $10\sqrt{3}\ cm^2$



Determine the **exact value** of;

(a) the acute angle θ

$$Area = \frac{1}{2}ab\sin C = 10\sqrt{3}$$

$$\Rightarrow \quad \frac{1}{2}.8.5.\sin C = 10\sqrt{3}$$

$$\Rightarrow \quad \sin C = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \quad C = \frac{\pi}{3}$$

accept $60°$

(b) the **area** of the shaded region

$$Area = 10\sqrt{3} - \frac{1}{2}r^2\theta$$

$$= 10\sqrt{3} - \frac{1}{2}.3^2.\frac{\pi}{3}$$

$$= 10\sqrt{3} - \frac{3\pi}{2}\ cm^2$$

Question 4 [8 marks – 1, 3, 4]

(a) Express 18° in radians..

$$18^\circ = \frac{18}{1} \cdot \frac{\pi}{180}$$
$$= \frac{\pi}{10}$$

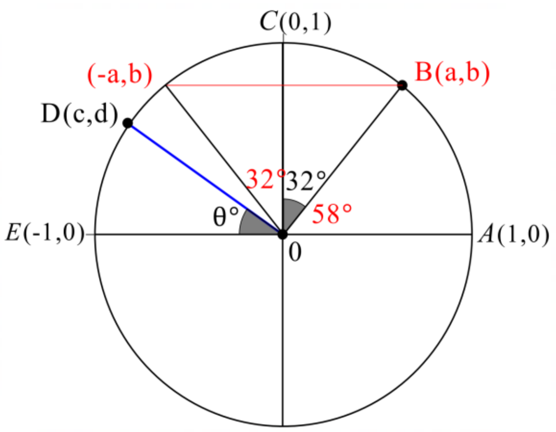
(b) Evaluate $\sqrt{3}\sin\left(\frac{2\pi}{3}\right) - \tan^2\left(\frac{5\pi}{6}\right)$

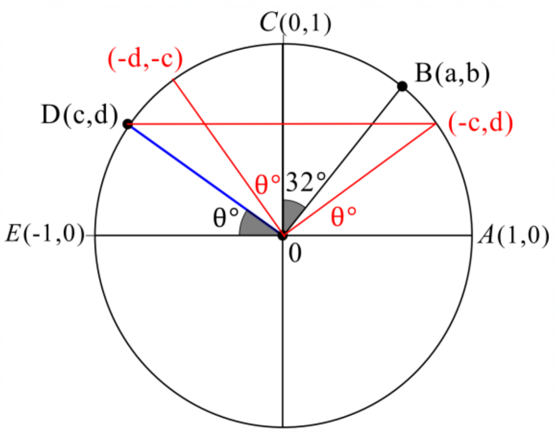
$$= \sqrt{3}\left(\frac{\sqrt{3}}{2}\right) - \left(-\frac{1}{\sqrt{3}}\right)^2$$
$$= \frac{3}{2} - \frac{1}{3}$$
$$= \frac{7}{6}$$

(c) Points $B(a,b)$ and $D(c,d)$ are plotted on the unit circle, where

$\angle COB = 32^\circ$ and $\angle DOE = \theta^\circ$

In terms of a, b, c and/or d, find ;





(i) $\sin 58^\circ$ $\quad b$

(ii) $\cos\theta$ $\quad -c$

(iii) $\tan 122^\circ$ $\quad -\frac{b}{a}$

(iv) $\cos(\theta + 90^\circ)$ $\quad -d$

Question 5 [3 marks]

Find the equation of the straight line through the point $(3, 5)$, which is perpendicular to the straight line passing through the points $(4, -6)$ and $(2, 3)$.

$$m = \frac{3-(-6)}{2-4}$$
$$= \frac{-9}{2}$$

required gradient $= \frac{2}{9}$

$$y = \frac{2}{9}\, x + c$$

$sub(3,5) \Rightarrow \quad 5 = \frac{2}{9}.3 + c$

$\Rightarrow \quad c = \frac{13}{3}$

$\therefore$ *required equation* $y = \frac{2}{9}x + \frac{13}{3}$ *or* $9y - 2x - 39 = 0$

Question 6 [5 marks- 2, 3]

The graph of $y = f(x) = -x^2 + x + k$ is drawn below.
The discriminant is equal to 25, ($\Delta = 25$).

(a) Show that the value of $k = 6$

$$\Delta = b^2 - 4ac = 25$$
$$\Rightarrow 1 - 4(-1)k = 25$$
$$\Rightarrow 4k = 24$$
$$\therefore \quad k = 6$$

last two lines must be shown

(b) Hence or otherwise, state the equation of the graph in the form $y = b(x-c)^2 + d$

$$y = -x^2 + x + 6$$
$$= -(x^2 - x \qquad -6)$$
$$= -(x^2 - x + \frac{1}{4} \qquad -\frac{1}{4} - 6)$$
$$= -[(x - \frac{1}{2})^2 \qquad - \frac{25}{4})$$
$$= -(x - \frac{1}{2})^2 + \frac{25}{4}$$

End of Calculator Free section

MATHEMATICS METHODS : UNITS 1 & 2, 2023

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1.2.1 to 1.2.8, 1.2.13

PM

Time Allowed	First Name	Surname	Marks
21 minutes			21 marks

Circle your Teacher's Name: Mrs Alvaro, Mr Bartoshefski, Mr Buckland, Mr Liu, Mr Lockyer, Mr Luzuk, Mr Mohammadi, Ms Narendranathan, Mr Ng, Mr Stephens, Mr Tanday

Assessment Conditions: ***(N.B. Sufficient working out must be shown to gain full marks)***

- Calculators: Allowed
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PART B – CALCULATOR ASSUMED

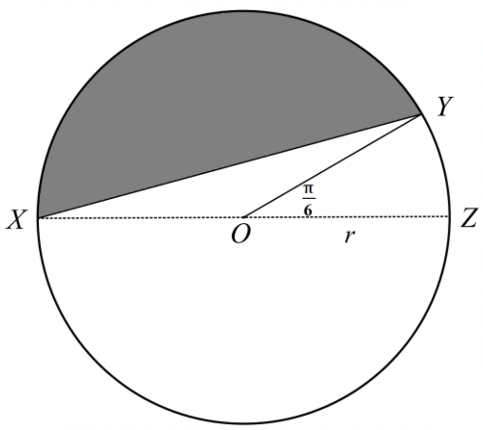
All approximate answers to be stated to an accuracy of 2 decimal places, unless otherwise directed

Question 7 [3 marks]

A circle with centre O , and radius r ,

has $\angle YOZ = \dfrac{\pi}{6}$

Determine the area of the shaded region if the radius is $10\,cm$.



$$Area = \frac{1}{2}r^2(\theta - \sin\theta)$$

$$= \frac{1}{2}10^2\left(\frac{5\pi}{6} - \sin\frac{5\pi}{6}\right)$$

$$= 50\left(\frac{5\pi}{6} - \frac{1}{2}\right)$$

$$= 105.90\ cm^2$$

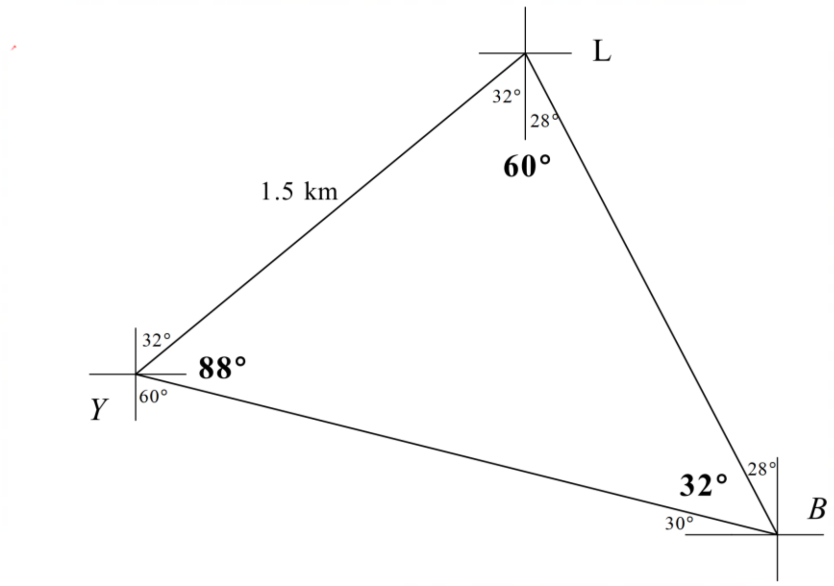
Question 8 [7 marks – 4,3]

A yacht is located at point Y and is sailing on a bearing $032° \; T$ towards a lighthouse at point L , $1.5 km$ from point Y .

From point Y , the yacht's navigator spots a boat at point B bearing $120° \; T$.

The bearing of the lighthouse from the boat at B is $332° \; T$.

(a) Draw a diagram and calculate the distance between the yacht and the boat.



$$\frac{\sin 60°}{x} = \frac{\sin 32°}{1.5}$$

$$\Rightarrow \quad x = 2\,4514$$

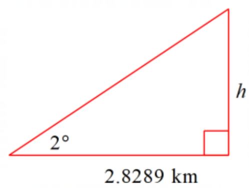
$$= \; 2.45 \, km$$

(b) From the boat, the top of the lighthouse has an elevation of $2°$. Determine the height (**to the nearest metre**) of the lighthouse above sea level.

$$y^2 = 1.5^2 + (2.4514)^2 - 2.(1.5)(2.4514)\cos 88°$$

$$\Rightarrow \quad x = 2.8289 \; km$$

$$Now \quad \tan 2° = \frac{h}{2.8289}$$



$$\Rightarrow \; h = 0.0988 \; km$$

$$= 99 \; m \; (to\,the\,nearest\,metre)$$

Question 9 [3 marks]

For the quadratic function $y = f(x)$, $f(2) = 0$ and $f(-4) = 0$.
If the minimum value is -6 , find the rule for $y = f(x)$

$$y = f(x) = a(x-2)(x+4)$$

$$sub(-1,-6) \Rightarrow \quad -6 = a(-1-2)(-1+4)$$

$$\Rightarrow \quad -6 = -9a$$

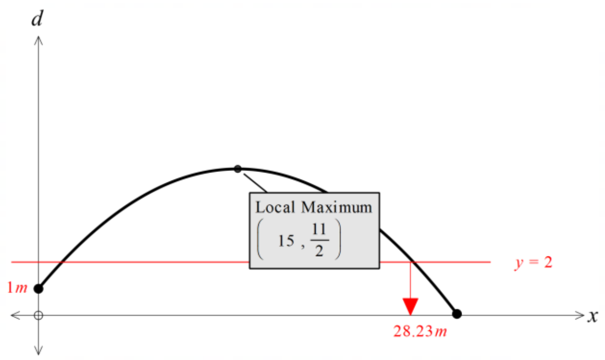
$$\therefore \quad a = \frac{2}{3}$$

$$\therefore \quad y = f(x) = \frac{2}{3}(x-2)(x+4) \quad or \quad y = \frac{2}{3}(x+1)^2 - 6$$

Question 10 [4 marks-1,1,2]

Ali, the cricketer, struck a ball such that its height, d metres , after it had travelled x metres horizontally, was given by the rule

$$d(x) = 1 + \frac{3}{5}x - \frac{1}{50}x^2 \quad , \quad x \geq 0$$



(a) At what height was the ball when it was struck ? 1 metre

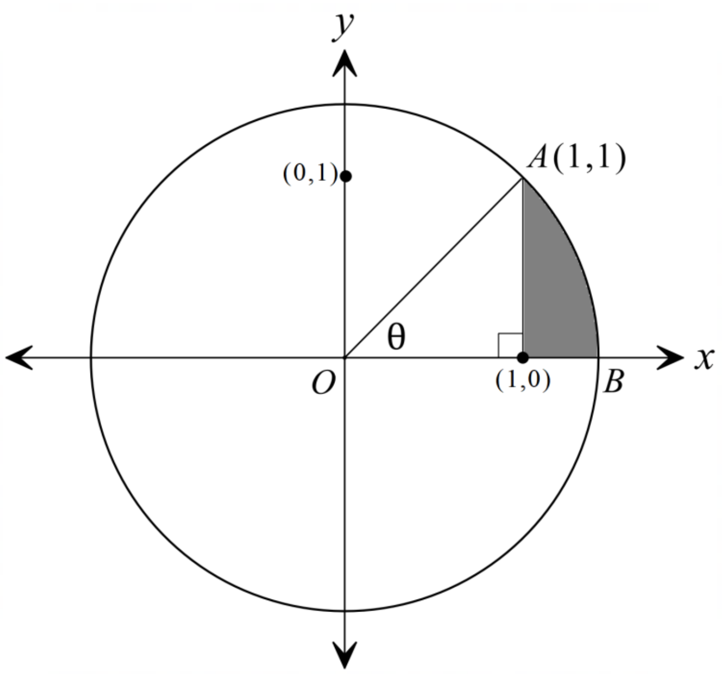
(b) What was the maximum height reached by the ball? 5.5 metres

(c) If the fielder, Sunny, who was more than 10 m away from batter, caught the ball when it was 2 m above ground, how far was Sunny from the batter?

28.23 metres

Question 11 [4 marks – 1, 2, 1]

Consider the circle drawn below, centred at O, the origin $(0,0)$.The line segment connecting O , and the point $A(1,1)$ makes an angle of θ with the positive x-axis



(a) Determine the coordinates of point B

$$OA^2 = 1^2 + 1^2$$
$$\Rightarrow \quad OA = \sqrt{2}$$
$$\therefore \quad B(\sqrt{2}, 0)$$

(b) Determine that the **exact length** of the minor arc AB

$$\tan\theta = 1$$
$$\Rightarrow \quad \theta = \frac{\pi}{4}$$

$$l = r\theta$$
$$= \sqrt{2} \cdot \frac{\pi}{4}$$
$$= \frac{\sqrt{2}\pi}{4} \text{ units}$$

(c) Hence, or otherwise, find the exact perimeter of the shaded region

$$Perimeter = 1 + (\sqrt{2} - 1) + \frac{\sqrt{2}\pi}{4}$$
$$= \sqrt{2} + \frac{\sqrt{2}\pi}{4} \text{ units}$$

End of Calculator Assumed section

Extra working page: